

NAVIN D

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Mechanical R&D Lead

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PROFESSIONAL EXPERIENCE

Tutr Hyperloop Private Limited

Chennai, India

Lead – Digital Solution 02/2026 – Present

- Leading the development of a Linear Induction Motor (LIM) Digital Twin for Hyperloop pod predictive maintenance, encompassing performance analysis.
- Delivered 2 real-time Digital Twin PoCs: (1) Structural Twin – 100 ms update, NI DAQ + MATLAB + FEA validation; (2) Battery Twin – Raspberry Pi + INA226 I²C + Simulink for live SOC/SOH estimation.
- Co-Authored and submitted a technical proposal for a ₹30 Crore Digital Twin initiative to Indian Railways.
- Established organization's first Digital Engineering capability; mentors one intern directly and guides a junior engineer.

Project Engineer 09/2024 – 01/2026

Promoted from CFD Engineer

- Led end-to-end execution of Tutr's first Hyperloop Top Hat program (₹20 lakh, 10-month), driving optimization, composite manufacturing, and cross-functional delivery to a successful minister-level technology showcase.
- Designed 3.0 m composite body in SOLIDWORKS, achieving Cd ≈ 0.25 via 20+ CFD iterations (ANSYS Fluent, k- ω SST) across 15+ conditions; validated via 1:30 scale prototype and Tesla Model S benchmark.
- Directed glass fiber hand lay-up (0°/45°/90° laminates); delivered 60 kg production assembly. Coordinated 5+ vendors and led integration; showcased to Hon'ble Railway Minister, Mar 2025.
- Developed Tutr's first composite vacuum chamber (₹22L, ~12 mo): Ø800 mm × 2 m carbon fiber + aluminum, achieving 1 kPa absolute with <5 kPa rise over ~18 hr; weight ~70 kg. Redesigned from first prototype failure using FEA-driven iterations (~15 cycles).

CFD Engineer 03/2023 – 08/2023

- Built external aerodynamic simulation capability in ANSYS Fluent (k- ω SST, hexa-core meshing, boundary-layer inflation).
- Benchmarked methodology against Tesla Model S before applying to Hyperloop body optimization.

PROFESSIONAL SUMMARY

Mechanical R&D Lead with 3+ years delivering first-of-kind systems from concept to deployment. Stepped up to own a ₹20L Hyperloop Top Hat program as CFD Engineer after a managerial transition, delivering from aerodynamic optimization through composite manufacturing to minister-level showcase. Mentors one intern directly and guides a junior engineer. Progressed from CFD Engineer to Lead in 36 months.

TECHNICAL SKILLS

Product & Design

SOLIDWORKS | DFM | Composite Structures (Glass/CFRP Hand Lay-up) | Vacuum Systems | System Integration

Simulation

ANSYS Fluent (CFD) | ANSYS Mechanical (FEA) | Aerodynamic Optimization | Structural Assessment

Digital Engineering

MATLAB | Simulink | App Designer | Python | Digital Twin Architecture

Instrumentation

NI USB-6003 DAQ | Raspberry Pi | INA101 | INA226 | I²C | Strain Gauges | Sensor Integration

KEY IMPACT METRICS

₹42L+ Budget Owned & Delivered	CFD → Lead in 36 Months with Team Growth
Cd 0.25 20+ Iterations First Top Hat	1 kPa 18-Hr Hold First Chamber

EDUCATION

M.E. Aerospace Engineering

Madras Institute of Technology (MIT), Anna University

2019 – 2021

B.E. Mechanical Engineering

Agni College of Technology, Anna University

2015 – 2019